

High dose rate and low dose rate ruthenium brachytherapy for uveal melanoma. No association with ocular outcome

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ABSTRACT

Background Episcleral brachytherapy is the most common eye-preserving option for treating uveal melanoma, with ruthenium-106 widely used in Europe. Case series have reported on ocular outcome, but there are very little data on the impact of dose rate. Therefore, we studied the association of dose rate with secondary enucleation.

Methods Data for all patients in Sweden managed with brachytherapy from 1979 to 2012 were retrieved (962 patients) and divided in quartiles based on dose rates at the tumour apex and at the scleral surface. Kaplan-Meier curves for secondary enucleation were generated independent of dose rate and for each dose rate quartile. Cox regression univariate and multivariate modelling included clinical parameters and dose rates at the tumour apex and the scleral surface.

Results Baseline clinical characteristics were not significantly different across the quartiles except that thinner tumours had higher dose rates at the tumour apex. Dose rates ranged widely, with a near normal distribution at the scleral surface but skewed at the tumour apex. For all quartiles, secondary enucleation was more often caused by tumour progression than by ocular side effects. Univariate and multivariate modelling including clinical parameters showed no association between dose rate and secondary enucleation.

Conclusions Although dose rates ranged widely, there was no apparent association with secondary enucleation, suggesting that based on ocular survival, current practice of annual replacement of applicators is acceptable, irrespective of the significant reduction in each applicator's dose rate. Nevertheless, patients treated with higher dose rates have shorter plaque exposure and presumably less ocular irritation.

INTRODUCTION

In the USA and Europe, episcleral brachytherapy is now used more often than enucleation of the eye to manage uveal melanoma.^{1,2} A range of radionuclides featuring different characteristics are available and in Europe ruthenium-106 is most widely used to treat uveal melanoma.^{2,3} Ruthenium brachytherapy was introduced in Sweden in 1979 using a standard dose of 100 Gy delivered to the tumour apex.⁴ Although this dose still conforms to that of most centres using ruthenium brachytherapy for uveal melanoma,³ the dose rate at the tumour apex is less frequently reported. Recently published guidelines for iodine-125 episcleral brachytherapy of uveal melanoma suggest the use of a minimum dose rate of 0.6 Gy/h at the tumour apex, but there are no similar guidelines for ruthenium

brachytherapy. Recent data suggest that high dose rate ruthenium brachytherapy may be associated with a lower metastatic rate and lower local recurrence rate when dose rate at the tumour apex exceeds 4 Gy/h.⁵ Results from other reports, however, are conflicting and have not been able to correlate local tumour recurrence with the tumour apex dose rate.^{6,7}

Ruthenium-106 is a β ray (electron) emitter with a half-life of 373.59 days meaning that dose rate is reduced to about half after 1 year.³ The nominal activity of commercially available ruthenium episcleral applicators ranges considerably between 4.1 MBq and 40.3 MBq (equal to 0.11 mCi to 1.09 mCi) at the time of delivery.³ Also, the tissue penetration of electrons is comparatively low and a dose rate of 6.6 Gy/h at a depth of 1 mm will drop to only 1.8 Gy/h at 5 mm depth.³ Typically, the prescription dose is set to 100 Gy at the tumour apex for a tumour less than 7 mm in height.^{2,3} When the tumour apex dose is kept constant, the apex dose rate will depend on tumour thickness; that is, because of low penetration thick tumours will be treated at a lower dose rate at the apex than thin tumours. Conversely, thick tumours will require a (much) higher scleral surface dose than thin tumours to deliver a similar dose at the apex. The time required to achieve the higher scleral dose necessary to treat a thick tumour will obviously be longer, but the scleral dose rate will be similar for a thick and a thin tumour. This indicates that the dose rates at the tumour apex and at the scleral surface should be analysed and correction for the association between tumour thickness and the tumour apex dose rate needs to be made. To study the effect of different dose rates on ocular outcome we retrospectively studied a large series of consecutive patients receiving ruthenium brachytherapy for uveal melanoma with univariate and multivariate statistical modelling.

PATIENTS AND METHODS

This study received ethics committee approval (2012/271-31/2; 07-03-2012) and adhered to the tenets of the Declaration of Helsinki. The computerised Medlog V2012-8 (Information Analysis, Crystal Bay, Nevada, USA) registry of all patients in Sweden with brachytherapy for uveal melanoma was searched for patients with ruthenium-106 brachytherapy and complete data including dose rates at the scleral surface and tumour apex. From the introduction of ruthenium brachytherapy in Sweden in 1979 up to and including 2012, a total of 1020 patients were treated and complete data



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Table 1 Clinical characteristics of tumours divided into quartiles based on dose rate at the tumour apex

	Dose rate 0.01–0.64 Gy/h* (n ² =242) Mean (SD)† (n)‡	Dose rate 0.65–1.09 Gy/h* (n ² =239) Mean (SD)† (n)‡	Dose rate 1.10–1.69 Gy/h* (n ² =242) Mean (SD)† (n)‡	Dose rate 1.70–4.76 Gy/h* (n ² =239) Mean (SD)† (n)‡
Tumour diameter (mm)	11.6 (3.2) (237)	10.4 (3.0) (236)	9.6 (3.0) (239)	8.2 (2.8) (236)
Tumour thickness (mm)	6.5 (1.2) (242)	5.0 (1.0) (239)	3.9 (0.9) (242)	2.8 (0.7) (239)
Distance to fovea (mm)	4.0 (4.0) (214)	3.3 (3.7) (221)	3.0 (2.9) (224)	2.9 (3.4) (219)
Distance to optic disc (mm)	4.5 (3.7) (218)	3.7 (3.3) (223)	3.6 (3.0) (230)	3.2 (3.3) (225)
Age (years)	60.9 (14.6) (242)	63.8 (13.8) (239)	63.6 (13.2) (242)	63.5 (14.1) (239)
Laterality (right:left)	130:112	116:123	144:98	119:120
Gender (men:women)	128:114	104:135	117:125	116:123

*Gray/hour.

†SD of the mean.

‡Number of patients.

were available for 962/1020 (94.3%). Patients not eligible because of missing data were studied for possible ascertainment bias. Throughout the period of study, all patients had treatment planned to include a delivery of 100 Gy to the prescription point of the tumour apex. Notably, iodine brachytherapy was not available in Sweden until 1999 and before then 39 tumours thicker than 7 mm were treated with ruthenium using less than 100 Gy to the tumour apex as an alternative to primary enucleation. All patients had the tumour thickness measured preoperatively by standardised A-scan ultrasonography. In 1999, the protocol for ruthenium brachytherapy was amended to allow an increase of the maximum scleral dose from 1000 Gy to 1500 Gy. In combination with the introduction of iodine brachytherapy, this effectively limited the use of ruthenium brachytherapy to tumours 7 mm or thinner. A subset of patients received preoperative (94 patients; 9.8%) or postoperative (134 patients; 14%) adjunctive argon laser photocoagulation or transpupillary thermotherapy. All radioactive plaques were commercially available (Eckert & Ziegler BEBIG GmbH, Berlin, Germany). Radioactivity data from the manufacturer were independently verified throughout the period of the study. The dosimetry used in this study was based on a method previously published.⁸ Surgery was usually performed in general anaesthesia and typically included transillumination followed by plaque positioning with a minimum 2 mm margin around the melanoma. Some tumours close to the fovea had eccentric plaque fixation and adjunctive laser therapy because of the compromised radiotherapy margin. Juxtapapillary melanomas were either

treated with a notched plaque or standard plaque combined with adjunctive laser therapy.

Ocular survival was measured from brachytherapy to posttra-diotherapy enucleation or end of follow-up. All statistical calculations were performed using the Statistica V.12 software (Statsoft Scandinavia AB, Uppsala, Sweden). To allow for multiple comparisons, the level of statistical significance was set at $p < 0.01$. Frequency distributions were studied by the median value and IQRs as frequencies were not normally distributed. Patients were divided in two sets of approximate quartiles based on the dose rates at the tumour apex and the scleral surface, respectively. Differences between baseline characteristics in the quartile groups were evaluated using the independent t test and one-way analysis of variance for continuous variables and the χ^2 test for categorical variables. Kaplan-Meier survival curves were generated for each dose rate quartile and compared with the logrank test. Cox regression modelling was used for univariate and multivariate analyses. Dose rate data were consistently treated linearly in this model as distribution was near-normal for dose rates at the scleral surface, although skewed for dose rates at the tumour apex.

RESULTS

The 58/1020 (5.7%) patients with missing dose rate data and not eligible for the study had similar baseline tumour and patient characteristics as included patients (not shown). Tumours with a low dose rate at the tumour apex tended to be thicker tumours ($p < 0.0001$). These tumours also tended to

Table 2 Clinical characteristics of tumours divided in quartiles based on dose rate at the scleral surface

	Dose rate 1.66–4.47 Gy/h* (n ² =241) Mean (SD)† (n)‡	Dose rate 4.48–5.66 Gy/h* (n ² =240) Mean (SD)† (n)‡	Dose rate 5.67–6.80 Gy/h* (n ² =239) Mean (SD)† (n)‡	Dose rate 6.81–11.11 Gy/h* (n ² =242) Mean (SD)† (n)‡
Tumour diameter (mm)	9.8 (3.3) (239)	9.8 (3.4) (233)	10.1 (3.2) (236)	10.0 (3.2) (240)
Tumour thickness (mm)	4.8 (1.9) (241)	4.3 (1.6) (240)	4.4 (1.6) (239)	4.6 (1.6) (242)
Distance to fovea (mm)	3.0 (3.0) (221)	3.5 (3.9) (217)	2.8 (3.1) (223)	3.8 (4.0) (217)
Distance to optic disc (mm)	3.6 (3.0) (227)	4.0 (3.6) (220)	3.3 (2.8) (224)	4.0 (3.9) (225)
Age (years)	62.7 (13.6) (241)	62.5 (14.5) (240)	62.8 (13.6) (239)	63.9 (14.2) (242)
Laterality (right:left)	123:118	146:94	118:121	122:120
Gender (men:women)	122:119	121:119	114:125	108:134

*Gray/hour.

†SD of the mean.

‡Number of patients.

Table 3 Tumour category distribution

Tumour category	Number	Per cent
T1A	409	42.5
T1B	10	1.0
T2A	416	43.2
T2B	8	0.8
T3A	88	9.1
T3B	1	0.1
T4A	16	1.7
No data	14	1.5

American Joint Committee on Cancer Classification (AJCC), 7th edition.

have a larger tumour diameter and a slightly longer distance to the fovea and optic disc ($p < 0.01$). This may be confounded as thicker tumours treated with a standard dose at the apex inherently will feature a lower dose rate at the apex. Thick tumours

would in turn be expected to have a larger than average tumour diameter (table 1). Although, thicker tumours were managed with a lower dose rate at the scleral surface ($p = 0.0014$), this difference was evident only when comparing the two lower dose rate quartiles ($p = 0.00056$). Also, tumours closer to the fovea were treated at a lower scleral surface dose rate ($p = 0.0077$), but on pairwise comparison the difference was only statistically significant between the two upper dose rate quartiles ($p = 0.0023$) (table 2). The large majority (86%) of tumours of included patients were category T1A and T2A (table 3).⁹

The frequency distribution showed that dose rates varied widely, at the scleral surface and at the tumour apex. The IQR of the tumour apex and scleral surface dose rates extended from 0.6 Gy/h to 1.7 Gy/h and from 4.5 Gy/h to 6.8 Gy/h, a 2.8-fold and 1.5-fold difference, respectively. The scleral surface dose rate showed a near-normal distribution, but the apex dose rate distribution was markedly skewed. This is consistent with a large proportion of thin tumours treated with a high dose rate at the tumour apex (figure 1A, B).

Figure 1 Frequency distribution of dose rates at (A) tumour apex and (B) scleral surface.

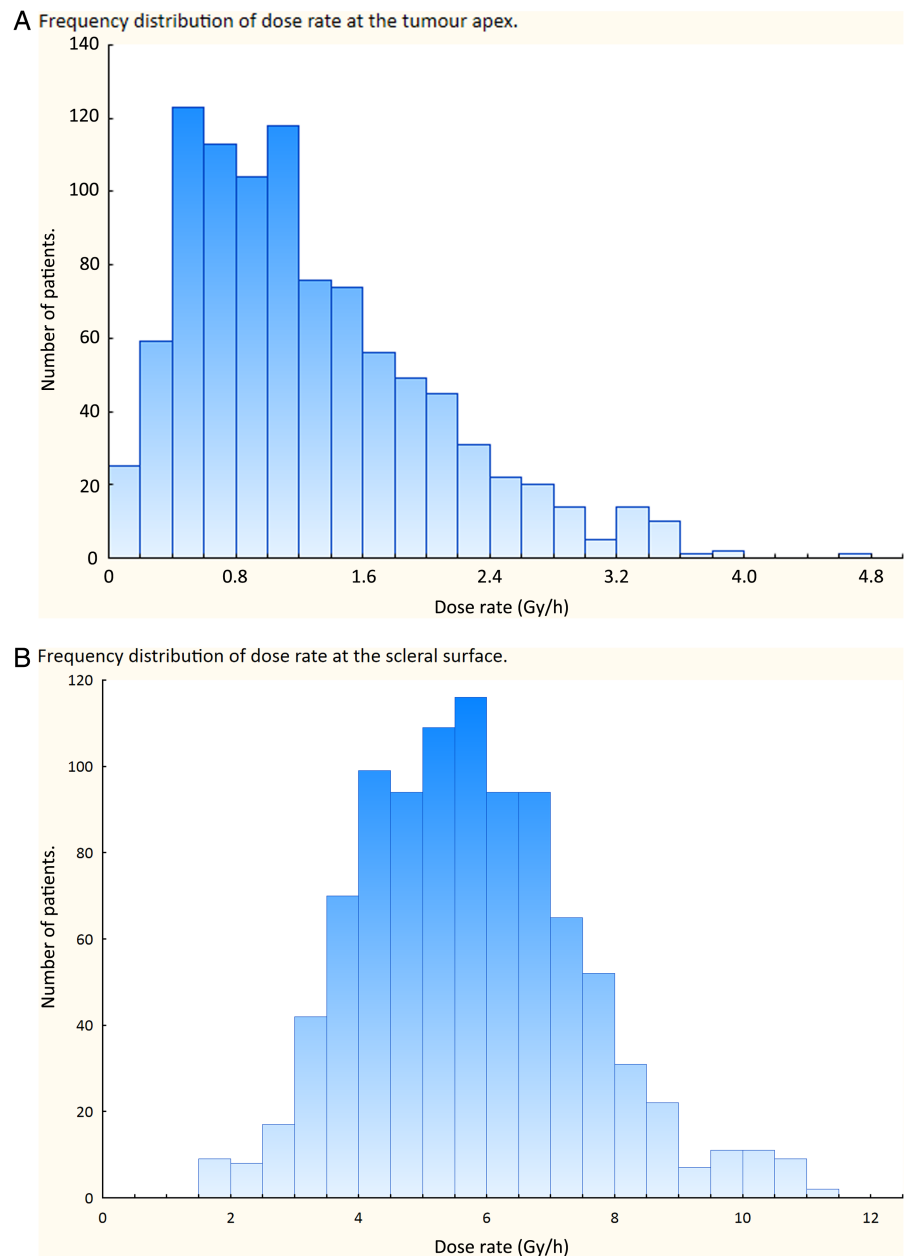
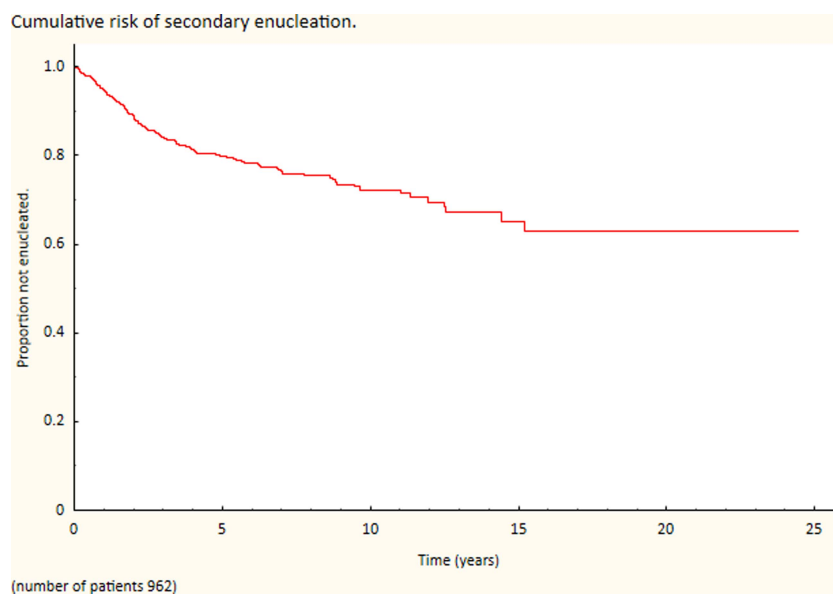


Figure 2 Kaplan-Meier survival curve of secondary enucleation (proportion of eyes retained) after ruthenium brachytherapy for uveal melanoma.



A total of 170/962 (18%) patients had secondary enucleation. Eye retention rate was 80% at 5 years and 72% at 10 years (figure 2). More eyes were enucleated after brachytherapy during the first few years, possibly because of a learning curve. Survival curves for ocular retention of eyes with tumours managed before and after the 1999 introduction of iodine brachytherapy and protocol amendment were, however, largely overlapping (not shown). The overall cause for secondary enucleation was local tumour progression in 75%. The remaining eyes were removed because of significant collateral ocular damage like neovascular glaucoma, extensive radiation retinopathy or optic neuropathy. Some eyes were blind but not all were painful and then enucleated due to patient's preference. There was a trend suggesting that tumours treated with a high dose rate at the apex more often were enucleated because of tumour progression, however this may be confounded by such tumours on average being smaller. Notably, no such trend was seen for the dose rate at the scleral surface (tables 4 and 5).

Kaplan-Meier survival curves were generated for each dose rate quartile at the tumour apex and scleral surface, respectively. These four curves of each of the two sets were largely overlapping and no statistical differences between curves were detected (figure 3A, B). Univariate and multivariate Cox regression model including clinical parameters and dose rates at the tumour apex and scleral surface failed to detect any association between the dose rates and secondary enucleation. Clinical parameters independently predictive of secondary enucleation were tumour diameter, tumour thickness and distance to the optic disc in the univariate analysis and only tumour diameter in the multivariate analysis (table 6). To address dose rate

multicollinearity we performed two additional multivariate regressions including either tumour apex dose rate or scleral surface dose rate. The dose rate did not reach statistical significance in any of these models.

DISCUSSION

Surprisingly little is known of the effect of dose rate either as measured at the tumour apex or at the sclera when using ruthenium brachytherapy for uveal melanoma. Minimal but conflicting data have been published; some suggest a better effect when using high dose rate brachytherapy,⁵ others failed to detect an obvious benefit.^{6,7} Mössböck *et al* retrospectively studied 45 patients divided in two apex dose rate groups (seemingly arbitrarily divided in <4 Gy/h or ≥4 Gy/h). Ocular complications were more frequent, but local tumour control rate was lower in patients treated with higher dose rates. These differences were, however, not statistically significant. Conversely, when Papageorgiou *et al* included 189 patients, there was no statistically significant difference in local control rate. Finally, Damato *et al* studied 458 patients but could not associate local treatment failure with dose rate. Although apex dose rate will be lower in thick tumours, only one study included a multivariate model.⁶ Only one of these studies was designed to primarily study the effect of dose rate, albeit in a very small number of patients.⁵

The apparent strength of our study was the large number of patients and hence events (secondary enucleation). This increases the power to detect a statistically significant difference. Dosimetry was consistent during the 33-year period of study. Protocol amendments were made in 1999 allowing for an increase in the maximum scleral dose from 1000 Gy to

Table 4 Proportion enucleated due to tumour progression. Groups divided in quartiles based on dose rate at the tumour apex

Dose rate (Gy/h)* min-max (n)†	0.01–0.64 (242)	0.65–1.09 (239)	1.10–1.69 (242)	1.70–4.76 (239)
Enucleated (n) (%)	48 (20)	47 (20)	36 (15)	39 (16)
Due to progress (n) (%)	30 (63)	35 (74)	27 (74)	35 (90)

*Gray/hour.

†Number of patients.

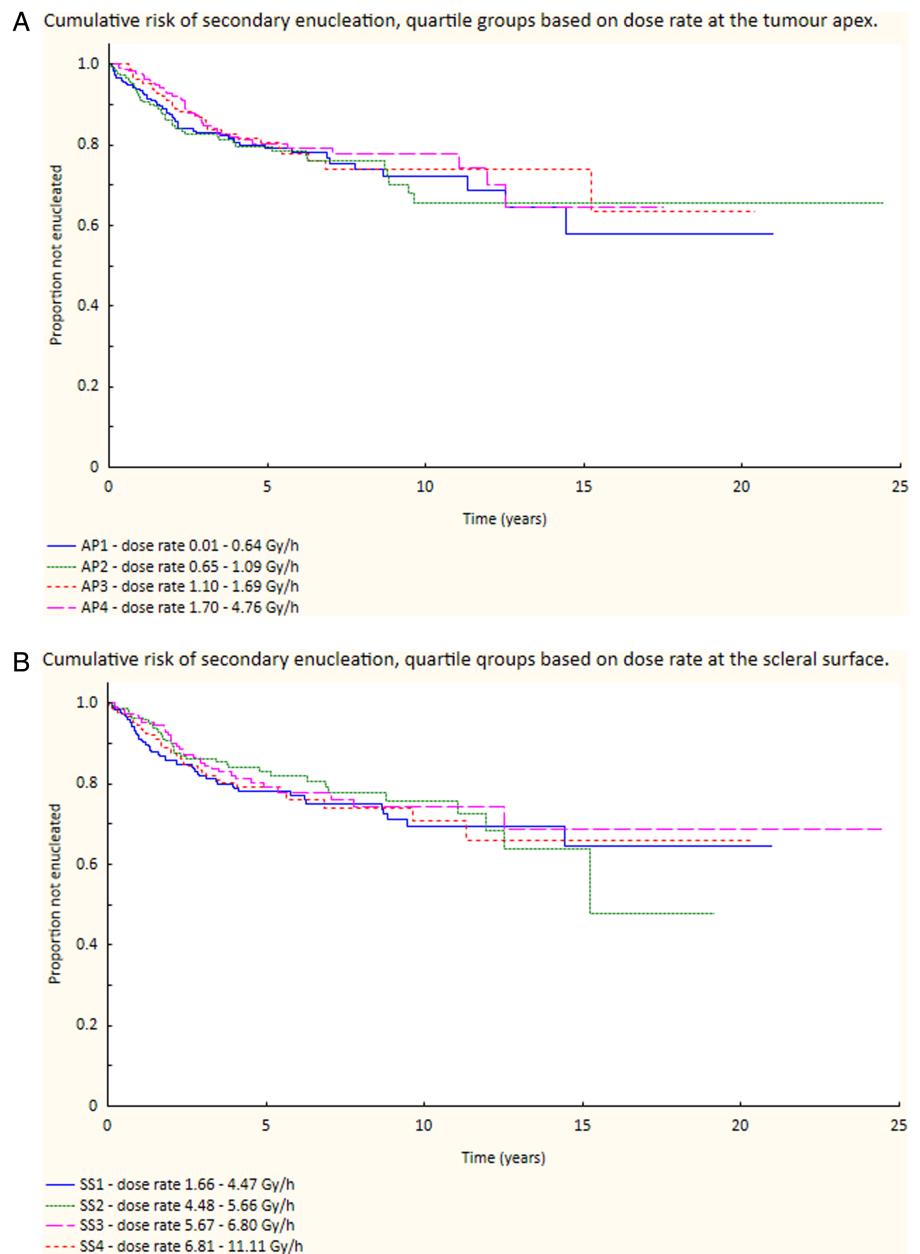
Table 5 Proportion enucleated due to tumour progression. Groups divided in quartiles based on dose rate at the scleral surface

Dose rate (Gy/h)* min-max (n)†	1.66–4.47 (241)	4.48–5.66 (240)	5.67–6.80 (239)	6.81–11.11 (242)
Enucleated (n) (%)	51 (21)	40 (17)	38 (16)	41 (17)
Due to progress (n) (%)	35 (69)	32 (80)	33 (87)	27 (66)

*Gray/hour.

†Number of patients.

Figure 3 Kaplan-Meier survival curves for quartiles of increasing dose rates at (A) tumour apex (AP1 (low)—AP4 (high)) and (B) scleral surface (SS1 (low)—SS4 (high)), respectively. The curves in each of the graphs are largely overlapping and there is no statistically significant difference in the rate of secondary enucleation between dose rate quartiles either at the tumour apex or at the scleral surface.



1500 Gy and exclusion of patients with tumours thicker than 7 mm (henceforth treated with iodine-125). The strategy before 1999 using ruthenium even for very thick tumours to avoid primary enucleation is not unique and a similar strategy has been used elsewhere.¹⁰ Taken collectively, however, we feel that these minor protocol changes are unlikely to significantly affect the results of this study, supported by largely overlapping survival curves for ocular retention of patients treated before and after 1999 (not shown). Importantly, this study was retrospective and as such potentially exposed to unrecognised bias. Indeed, patients were chosen for ruthenium brachytherapy but selection was based on near consistent criteria and patients were from all parts of Sweden.

Dosimetry for ruthenium brachytherapy is complicated and contentious using different standards.^{11 12} In 2002, the manufacturer BEBIG switched from the original ASMW (Amt für Standardisierung, Messwesen und Warenprüfung of the German Democratic Republic) specification (ASMW was the standardisation office of the former German Democratic

Republic) to the National Institute for Standards and Technology specification. The National Institute for Standards and Technology standard is believed to considerably reduce measurement uncertainty.³ To adhere to our own quality assurance system, the manufacturer's specifications have consistently been validated by our own independent measurements throughout the entire period of this study.^{8 13} The technique we adopted has been previously published and was used for treatment planning when discrepancies arose.⁸

Ruthenium-106 is a β ray emitter with a half-life of 373 days; that is, the dose rate at any given prescription point will be reduced to about half that of the original dose rate after 1 year of use.³ We replace our ruthenium plaques on an annual basis. Theoretically, this means that if tumour thickness was associated with the time of year treatment took place, this study would have been flawed. This would have been a surprise finding, but we did, unsuccessfully, look for any such pattern. Rather, the distribution appeared random across the year (not shown). Although this was a retrospective, non-randomised study, we

Table 6 Univariate and multivariate Cox regression analysis of secondary enucleation

Variable	Scoring	Univariate			Multivariate		
		p Value	HR	95% CI*	p Value	HR	95% CI*
Dose rate at the tumour apex (Gy/h)†	Value in Gy/h	0.67	1.02	0.94 to 1.1	0.69	0.92	0.60 to 1.40
Dose rate at the scleral surface (Gy/h)	Value in Gy/h	0.11	0.85	0.70 to 1.04	0.34	1.05	0.94 to 1.17
Tumour diameter (mm)	Value in mm	0.0018	1.08	1.03 to 1.13	0.049	1.06	1.00 to 1.12
Tumour thickness (mm)	Value in mm	0.018	1.11	1.02 to 1.21	0.79	1.03	0.85 to 1.24
Tumour distance fovea (mm)	Value in mm	0.062	0.95	0.90 to 1.00	0.87	0.99	0.92 to 1.08
Tumour distance optic disc (mm)	Value in mm	0.024	0.94	0.88 to 0.99	0.070	0.92	0.84 to 1.01
Laterality	Right=1, left=2	0.62	1.08	0.80 to 1.46	0.25	1.20	0.88 to 1.65
Age	Value in years	0.91	1.00	0.99 to 1.01	0.91	1.00	0.99 to 1.01
Gender	Male=1, Female=2	0.39	1.14	0.84 to 1.54	0.96	1.00	0.74 to 1.38

*CI of the estimated HR.
†Gray/hour.

therefore postulated that the scleral surface dose rate would be random for any clinical parameter. This was consistent with our findings of no statistically significant difference for any of the clinical parameters studied between the scleral dose rate quartiles. Conversely, and as would be expected, more tumours treated with a low dose rate brachytherapy at the melanoma apex tended to be thick. We therefore analysed our data set based on the scleral surface and the tumour apex dose rates, albeit both of these studies eventually failed to associate dose rate with secondary enucleation following brachytherapy.

Enucleation of the eye following brachytherapy is a binary, clearly identifiable event and we used this as our primary outcome measure, although the cause for enucleation differed between patients. The rationale could, however, in almost all cases be identified from the registry, albeit there was sometimes doubt as to whether the actual cause for enucleation was radiation retinopathy, optic neuropathy and/or simply the patient's preference when experiencing a blind eye. Tumour recurrence was identified solely by clinical examination and ultrasonography. Notably, clinical misdiagnosis has been reported after ruthenium brachytherapy with intratumoural haemorrhage being interpreted as tumour growth.¹⁴ Interestingly, recent data suggest that histopathological examination fails to verify the clinical findings of local tumour recurrence in 40% of eyes enucleated after iodine brachytherapy for uveal melanoma.¹⁵ This report included many eyes enucleated at smaller centres where the clinical misdiagnosis rate may be higher. Alternatively, a small area with tumour growth may possibly have remained undetected even by a meticulous histopathological examination. Either way, this raises the possibility that some of the eyes enucleated on clinical evidence of tumour growth may have been misdiagnosed.

Brachytherapy is currently employed in managing a number of non-ocular cancers and one clear advantage of high dose rate brachytherapy is to reduce treatment time and hence minimise exposure to staff. This is of far less concern in ruthenium brachytherapy as the penetration of radiation in this setting is low. Randomised trials or meta-analyses comparing outcomes after different dose rate strategies are notably scarce in non-ocular cancer. One recent meta-analysis of brachytherapy in oral cancer, however, suggests that local recurrence rate is similar after high versus low dose rate brachytherapy.¹⁶

Similarly, our results suggest that the dose rate in ruthenium brachytherapy for uveal melanoma, even within the very wide

range used in clinical practice, has little, if any, effect on ocular survival. Unfortunately, our study does not generate results from which clear guidelines for an appropriate dose rate range can be inferred. Nevertheless, a low dose rate will generate a long treatment time, likely to be paralleled with more ocular irritation.

Competing interests None.

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Provenance and peer review Not commissioned; externally peer reviewed.

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